

A Longitudinal Study of the Prevalence of WiFi Bottlenecks in Home Access Networks

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Abstract

Although home wireless networks (WiFi) are increasingly becoming performance bottlenecks, there are no research studies based on long-running field deployments that document this phenomenon. Given both public and private investment in broadband Internet infrastructure, a rigorous study of this phenomenon—and accompanying public data, based on open-source methods, is critical. To this end, this study pioneers a system and measurement technique to directly assess WiFi and access network performance. This study is first to continuously and contemporaneously measure Internet performance along two segments—the wireless client to the access point, and from the access point to the ISP access network. It is also the largest and longest-running study of its kind, with public data spanning more than two years (and counting), and, to our knowledge, the first such study in nearly a decade. Our study is based on data from over 22,000 joint measurements across more than 50 broadband access networks. Our findings have important implications for both the development of access technologies and Internet policy. Notably, for users with access links that exceed 800 Mbps, the user’s wireless network was the performance bottleneck 100% of the time. Such inflection points will continue to evolve, yet the contributions of this paper include not only the results, but also open-source tools, data, and ongoing continuous measurements.

CCS Concepts

• **Networks** → **Network performance analysis**; *Home networks*.

Keywords

WiFi Bottlenecks, Network Performance, Access Networks, Home Networks, Internet Access Performance, Network Traffic Analysis

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1 Introduction

Home Internet users invest significant resources and expense improving the state of their wired Internet access connections, which in the United States, now routinely exceed 1 Gbps for many service plans. As wireline broadband access Internet speeds continue to increase, however, home Internet users are increasingly faced with a new performance bottleneck: their home wireless network. The prospect of the home wireless network as an end-to-end performance bottleneck is, in fact, not new: nearly a decade ago, researchers noted that WiFi networks in homes routinely introduced throughput bottlenecks on home networks when the fixed-line ISP throughput exceeded 35 Mbps [16]. Wireless technologies have improved substantially over the last decade, naturally, yet fixed-line Internet throughput *also* has continued to increase, warranting a re-appraisal of this fundamental question.

The stakes for understanding home Internet performance bottlenecks have perhaps never been higher, as federal funding is poised to invest in Internet infrastructure across the United States. In many cases, Internet speed measurements are conducted with crowdsourced Internet measurement tools (e.g., Ookla’s SpeedTest, Measurement Labs’s NDT), and previous work has observed that client-based speed tests can often be the bottleneck in the end-to-end performance [4].

Indeed, Internet infrastructure does not stop at the “wall jack”, and the user experience often depends as much on infrastructure *inside* the home as it does on infrastructure *to* the home. It does little practical good for an Internet service provider to deliver fast speeds over a wired connection to a single point in the home if the wireless connectivity that users rely on in the rest of the home is poor. In April 2021, a Pew Research Center Survey found that “roughly half of [U.S.] broadband users report[ed] that they often (12%) or sometimes (37%) experienced problems with the speed, reliability or quality of their high-speed Internet connections at home” and that “29% of broadband users did something to improve the speed, reliability or quality of their [home Internet] since the beginning of the pandemic” [8]. Many people experience problems with their home Internet performance and want to improve it.

In this paper, we revisit the question of whether home wireless connectivity is a performance bottleneck in the modern era.

This paper makes the following contributions:

- We develop the first method to *jointly* and *continuously* measure the performance of the wired and wireless networks in a home network, using a combination of (1) a router-based speedtest performing measurements over the wide-area from the user’s wired network access point; (2) a browser-based plug-in that performs continual throughput measurements

from a user’s wireless device inside the home to a speedtest server running inside the home network at the router.

- We affirm and amend the presence of wireless performance bottlenecks in the home network, finding that the prevalence of WiFi bottlenecks increases significantly for homes with access speeds exceeding 100 Mbps, and that homes whose ISP speed tiers exceed 800 Mbps, the home WiFi network was the bottleneck in nearly 100% of the measurements we performed.
- We publicly release our measurement toolkit—router-based speedtest, browser-based home network speedtest, data, and associated analysis—to encourage the continued reassessment of these results, in a broader range of settings, and over time as ISP access networks and WiFi technologies continue to evolve.

2 Background and Related Work

In this section, we provide background on WiFi measurements in access networks, as well as previous work that has performed measurements of WiFi in different capacities and contexts.

2.1 Background

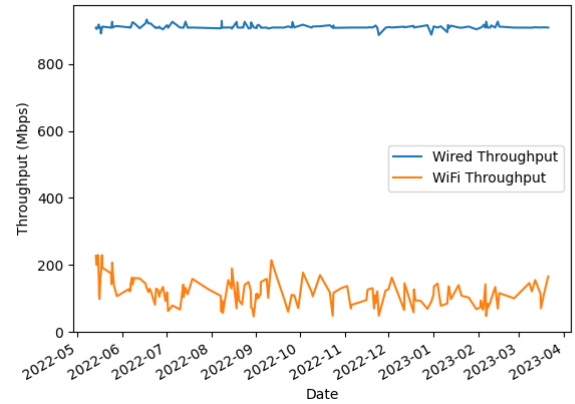
Wireless Internet access via the 802.11 protocol (“WiFi”) has become increasingly prevalent in recent years. As of 2021, 93% of households in the United States had an Internet connection, and the vast majority of these households rely on WiFi for connectivity within the home [10]. The number of devices connected to the Internet has also increased dramatically. As of 2020, the average U.S. household has 25 Internet-connected devices, more than double the number of connected devices in 2019 [3]. Hence, it has become increasingly important to understand whether, and to what extent, home wireless Internet connectivity affects the performance that users ultimately experience in their homes.

WiFi bottlenecks can significantly affect a user’s online experience. The access ISP connection might support high speeds, but if the corresponding WiFi connection is poor, a user will ultimately experience the network speeds corresponding to the slower WiFi connection. Significant attention (and funding) [2] is currently being devoted to improving Internet infrastructure. Ultimately, if performance bottlenecks are in home network infrastructure, investments must also be directed towards assessing and improving the WiFi equipment that households use to access the Internet.

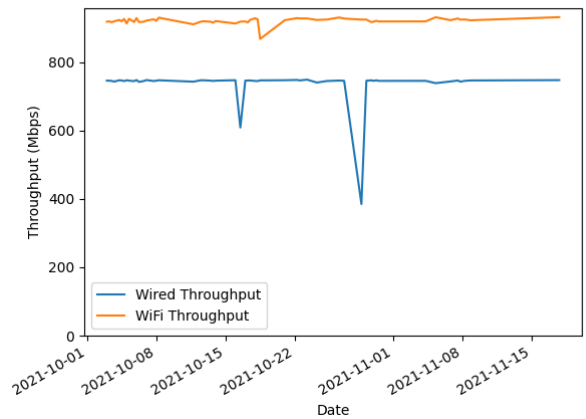
Figure 1 shows WiFi and access speeds for two households in a large city in the United States. Figure 1a has WiFi speeds below wired access speeds. WiFi is a bottleneck 100% of the time for this household. Conversely, in Figure 1b, WiFi speed exceeds access speed. Note that the household with WiFi bottlenecks is paying for Internet service with high access speeds (about 900 Mbps) whereas the household with no bottlenecks is paying for slightly lower access speeds (some of the households in our study have slower access speeds than this example).

2.2 Related Work

Various previous works have attempted to characterize wireless network performance in specific settings, including home networks. Early seminal work from Sundaresan et al. [16] characterized the



(a) WiFi bottleneck.



(b) Access bottleneck.

Figure 1: WiFi and wired bottlenecks in households in a large city in the United States over the past 21 months.

performance of 802.11b networks in a home setting, and found that when access link speeds exceeded approximately 30 Mbps, the home wireless network became the bottleneck link. This study’s measurement techniques were indirect: the technique used inferred the presence of bottlenecks based on inter-packet arrivals downstream of the wireless access point. In contrast, this work measures wireless network performance *directly*. Additionally, this previous work was conducted nearly a decade ago; this work thus constitutes a re-appraisal of this past work.

Previous work from Shi et al. [12] and Chung et al. [1] have also collected client-side wireless network measurements. However, these studies do not compare wireless measurements to wired measurements, as ours does. Sui et al. [14] also design and deploy a framework for collecting WiFi measurements. However, their framework measures and characterizes WiFi latency, while ours studies WiFi throughput.

More recently, Feamster and Livingood highlighted the challenges with performing client-based speed tests [4], and presented evidence

that in many cases, home wireless access links are the bottleneck in end-to-end Internet speed tests. Their work, however, was a survey of the shortcomings of client-based speed tests, of which wireless bottlenecks are one example. This work focuses specifically on the prevalence of wireless bottlenecks in home networks.

Many other studies have studied the prevalence of wireless performance in other contexts, although they have not explicitly focused on the extent to which poor WiFi performance introduces a bottleneck along an end-to-end path. Yang et al. developed a machine learning-based approach for predicting performance degradation in WiFi networks, showcasing the potential of predictive analytics in managing network performance [18]. Sui et al. investigated the impact of interference on WiFi performance, emphasizing mitigation strategies [13]. Sui et al. also developed mechanisms for characterizing and improving WiFi latency in large-scale operational networks [15]. Other work has explored WiFi performance in high-density environments [7], while other work has explored the implications of security mechanisms on WiFi performance [6]. In the context of the Internet of Things (IoT), Sheth et al. investigated how device heterogeneity contributed to bottlenecks in WiFi networks [11]. Other work has explored the effects of Quality of Service (QoS) mechanisms on WiFi bottleneck identification [17]. Recent work from Grazia explored the impact of emerging technologies, such as Wi-Fi 6, on TCP bottlenecks [5].

3 Method

Measurements: Netrics. We used the public, open-source Netrics platform [9] to conduct periodic downstream throughput measurements for both the access ISP connection and for the internal WiFi connectivity. Netrics is a widely accepted measurement platform, now in use across the United States for many studies of access Internet performance, that has been underway for over two years. It currently provides many open-source network measurement tests, including a variety of continuous throughput and latency tests. A common way of performing Netrics measurements is by distributing a single-board computer (typically, a Raspberry Pi) to study participants at vantage points of interest. The participants connect the Pis directly to routers or modems in their home networks, and the Pis measure and archive these performance metrics. The Netrics software distribution can schedule any network measurement and is commonly deployed to run three different periodic throughput tests: Measurement Labs’s NDT7, Ookla’s Speedtest, and iperf3 to a nearby server. The measurement suite also runs a variety of other measurements (e.g., DNS lookup latency, packet loss rates) that we do not analyze.

Contemporaneous measurements of WiFi and ISP. Directly comparing the throughput of the (typically wired) access ISP network to that of the WiFi network from the same household is extremely challenging. It requires deploying infrastructure in the home at the router both to conduct throughput measurements of the access ISP and to host a server for conducting wireless throughput measurements; it requires deploying software on one or more client devices within the home network to measure the home WiFi; and, finally, it requires collecting contemporaneous measurements of both the access ISP throughput and the home wireless network. To perform the throughput measurements from the home wireless network,

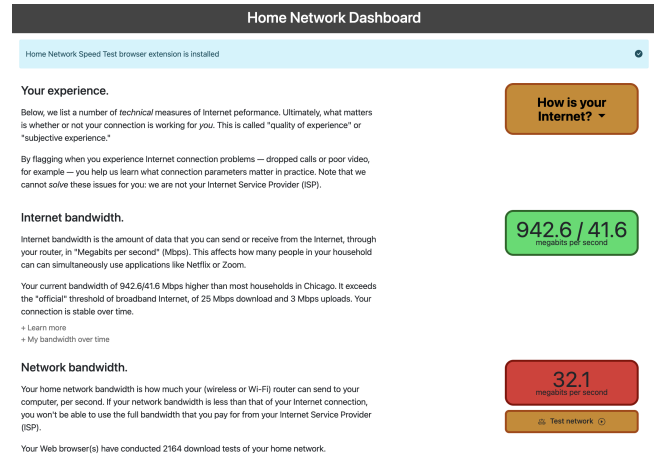


Figure 2: Netrics dashboard at <https://netrics.local/>.

we hosted a speed test server on a Raspberry Pi, intended to be deployed at or near the home router. A user could then direct his or her browser to the server at <https://netrics.local/>, which would trigger the browser to run a Javascript-based NDT7 throughput test to the server running on the Raspberry Pi. To enable *continuous* measurements, we also implemented a public, open-source Chrome browser plugin that would run periodic NDT7 tests from the user’s browser to the server running on the Raspberry Pi, even if the user didn’t explicitly initiate a test. We designed these throughput tests to run every few hours, but the actual frequency of the tests depended on whether someone was using the device on which the browser plugin was installed. Figure 2 shows an example of the public, open-source Netrics dashboard, which displays the results of the contemporaneous measurements of the access ISP and the home WiFi network. This particular example shows a case where the measured ISP access throughput (shown in green) is far greater than the measured WiFi throughput (shown in red), indicating that WiFi is the bottleneck.

Participant recruitment. We installed Netrics on a fleet of Raspberry Pis and recruited participants through community-based organizations and social media channels. Participants were offered a monetary incentive of \$100 to install a device to an access point for one month, an additional \$25 to continue hosting the device for another six months, and no incentives after that (akin to a “RIPE Atlas model” of ongoing deployment). We have been collecting data from the deployment since June 2021 among our research team, and since October 2021 across a broader set of study participants across one of the largest cities in the United States.

Data collection. Table 1 summarizes the measurements for the study. The measurements collected for this study were collected from September 18, 2021 through September 30, 2023¹. To measure download throughputs of both the access link (the access speed) and the wireless network (the WiFi speed), we use Measurement Lab’s Network Diagnostic Tool (NDT7) tool. We performed access throughput measurements for each household approximately once per hour. WiFi speed measurements did not follow a fixed schedule,

¹ Since consumer WiFi has not experienced significant changes from Septmeber 2023, we don’t expect the conclusions of this paper to change. All of our data is and will continue to be public. We continue to collect and analyze more recent data.

Period	September 18, 2021 - September 30, 2023
Households	52
Vantage Points	65
- < 50	3
- 50-100	15
- 100-200	3
- 200-400	10
- 400-800	21
- > 800	13
Measurements (Mbps)	22,363

Table 1: Summary of the data collected from the Netrics deployment.

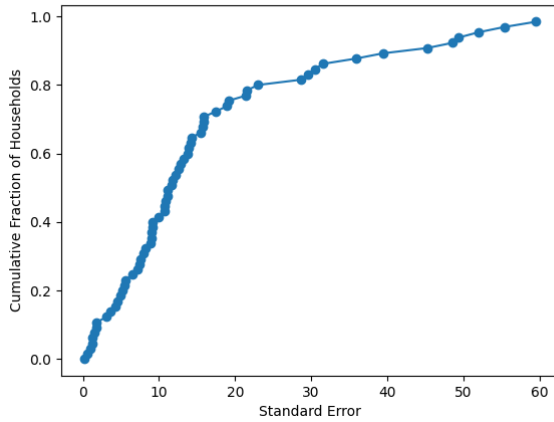


Figure 3: Sample Error for the Difference in WiFi and NDT Values per Household

due to the fact that the WiFi measurements were taken from a user's browser on a wireless device in the home and thus depended on the user actively using the device where the browser plug-in was installed. To ensure that the measurements compared were taken at around the same, we re-sampled the data for each household. We used a 6-hour time window and aggregated using the median access and WiFi speed within each window. We ran a time series analysis on windows of 24 hours, 18 hours, 12 hours, 6 hours, 3 hours, and 1 hour. We used a 6-hour time window because we found that this was the time period in which there was the most samples. This time window had a broad enough window to contain a large amount of data, yet small enough intervals to sufficiently splice the data.

Survey study. We completed a study via Qualtrics to better understand the WiFi setup in participants' homes. We asked participants which ISP provides their wired Internet connection, if they have WiFi extenders in their homes, and what their modem and router setup is. Most importantly, we asked participants to upload a PDF or screenshot of their most recent Internet bill to understand which speed tier they subscribe to. We use this speed tier data in our analysis.

Data analysis. We only want to include NDT7 measurements in our analysis because the WiFi data is measured using NDT7. Thus, we restrict our sample to that measurement tool to ensure that we are comparing comparable measurements of the link capacity and the

WiFi throughput for each household. We also restricted our sample of households to those that had a significant number of WiFi measurements during the analysis period, dropping households that did not have an enough WiFi measurements to allow us to draw strong conclusions about the performance of the wireless network and corresponding wired access network. After resampling the data for each household, we dropped households from the analysis that had less than 20 measurements with both WiFi speed and access speed. To determine this cutoff, we statistically found the sample error for the difference between WiFi and Internet speed measurements per household. Figure 3 is the CDF of sample error that we created based on these findings, through which we found that the cutoff must be 20 measurements. After dropping households from the dataset, our sample had 52 households, with an average of 272 WiFi and access speed coincident wireless and wired measurements per household.

Some households experienced a significant change in access speeds during the period of analysis, likely due to a change in Internet service provider or service plan. This occurred for 13 households in our sample. Because these changes in access speeds may significantly affect our analysis (i.e., by changing the wired throughput bottleneck for a household), we split the data for these households and treat each as a separate observation point. This ensured that the access speed is constant during the analysis period. As a result, we add 13 observation points to our sample, bringing the total number of observation points to 65.

Limitations. In our study, we lack insight into the types of devices that study participants use. Furthermore, we do not have complete visibility into the environment that the measurements are taken in. This does not affect our overall conclusions but bounds the types of conclusions that can be made. Our conclusions are still valid in a broader context—we identify that bottlenecks exist but do not know what they are directly attributed to. We are unable to draw conclusions about whether the bottleneck is generated by properties of the network or hardware. Additionally, our study primarily takes place in the Chicago area. However, we do not consider this to be a limitation that affects our conclusions because characteristics of WiFi are unlikely to vary by geography. Finally, our setup does not allow us to collect information about the WiFi standard. This does not affect our findings though, because our work is not concerned with evaluating WiFi standards. Rather, we report on the lived experiences of users with WiFi. Integrating information about the WiFi standard could be a topic for future investigation.

4 Results

To evaluate performance bottlenecks at the household level, we examine throughput test data from the Netrics devices across the deployment for the 24-month period of the study. After measuring the download throughputs of the WiFi and access connections, we compare the WiFi throughput of each household to its downstream access throughput over time, paying particular attention to the throughput of the WiFi network relative to corresponding ISP access network.

4.1 Prevalence of WiFi Bottlenecks

Figure 4 highlights the prevalence of WiFi bottlenecks across households in our study. To calculate the prevalence for each household,

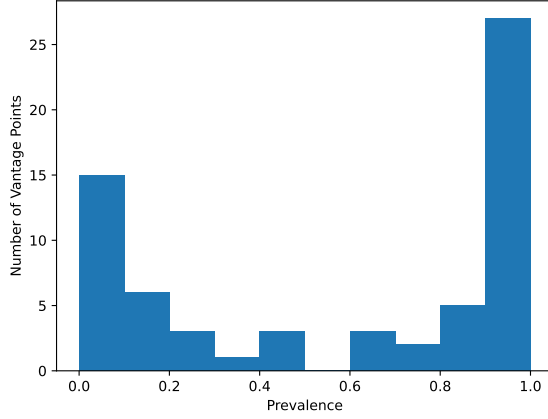


Figure 4: Prevalence of WiFi bottlenecks by vantage point.

we compare the household’s access throughput to its WiFi speed over time and note extent to which the WiFi throughput is lower than the access throughput. For example, if a household has 100 measurements of WiFi and access throughputs and, in 20 of 100 measurements, WiFi throughput is below access speed, then the bottleneck prevalence for this household would be 0.2; 89% of households experienced at least one WiFi bottleneck. This indicates that there was at least one measurement collected where WiFi throughput was below the access throughput. Across vantage points, the vast majority either experience no WiFi bottlenecks or a high percentage, as Figure 4 shows. Most vantage points do not have a prevalence between 10% and 80%, indicating in most cases, either WiFi is almost never a bottleneck, or it is almost always a bottleneck. As we will see, the circumstance depends on the household’s access speed tier.

Figure 5 shows the prevalence of WiFi bottlenecks across vantage points, according to the speed tier of the access network. Each point on each curve represents a vantage point; the x-axis represents the prevalence of WiFi bottlenecks at that vantage point, and the y-axis represents is the cumulative distribution of vantage points. As expected, vantage points with higher speed tiers see a greater prevalence of WiFi bottlenecks. For vantage points with access speed tiers of less than 100 Mbps, the prevalence of WiFi bottlenecks, tends to be less, with the median vantage point seeing a prevalence of less than 0.1. On the other hand, for vantage points whose speed tier exceeds 800 Mbps, the prevalence of WiFi bottlenecks is 1, indicating that *all* joint measurements in those households indicate the presence of a WiFi bottleneck.

This variation of the prevalence of WiFi bottlenecks also correlates with the measured throughput of the access ISP. The median access ISP throughput among households that experience frequent bottlenecks is 689.64 Mbps; on the other hand, the median access ISP throughput is only 58.35 Mbps for households that rarely experience bottlenecks. We find that WiFi bottlenecks become increasingly prevalent as measured access throughputs increase.

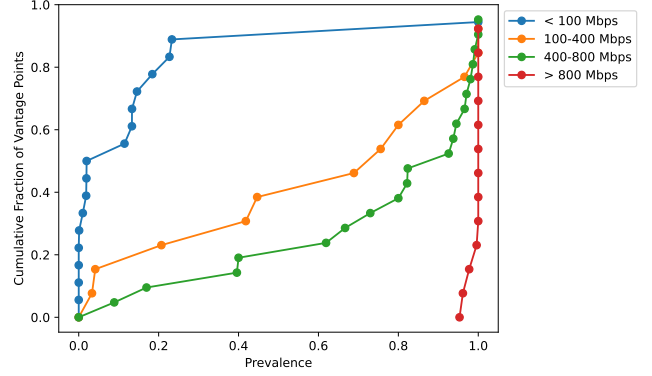


Figure 5: Prevalence of WiFi bottlenecks by speed tier.

4.2 Magnitude of WiFi Bottlenecks

To evaluate the magnitude of a WiFi bottleneck (i.e., how much slower the WiFi link is than the ISP access link), we compare the median WiFi throughput, access throughput, and the effective throughput of a household, which we define as the minimum of the median WiFi throughput and the median access ISP throughput.

The extent of the difference between the actual throughput and the access speed for each tier indicates the magnitude of the WiFi bottlenecks. For example, we find that households with speed tiers in the 200–400 Mbps range experience a mean actual throughput of 155.99 Mbps, while the mean access throughput is 265.60 Mbps. This disparity shows that, while paying for access throughputs between 200 and 400 Mbps, the median household is typically experiencing actual speeds that are 109.91 Mbps lower. Interestingly, this gap increases with higher throughput tiers. For households in the 400–800 Mbps tier, the gap between access throughput and actual speed grows to 240 Mbps, and for households in the 800 Mbps plus tier to 600.3 Mbps. This result indicates that home wireless networks introduce bottlenecks that often result in households failing to completely utilize the access throughput that they have purchased from their ISP.

5 Conclusion

This paper presents a first-of-its-kind study of direct and contemporaneous measurements of access throughput and WiFi throughput in the same home broadband network. It is the first such study in nearly a decade, constituting a reappraisal of WiFi’s role as a bottleneck in home broadband networks. The results are perhaps unsurprising, given past research highlighting WiFi bottlenecks in home networks and anecdotal folklore to this effect. The contribution of this paper is thus a formal confirmation of this effect, through measurement, open-source tools and data, and ongoing measurements that have, and continue to, track the evolution of this phenomenon as WiFi and access network technology evolve. We hope and expect that the tools and data that we release in conjunction with this paper will encourage ongoing research in this area, as well as applications to questions in other domains, including public policy.

Acknowledgments

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Appendix

Ethics. This work does not raise any ethical issues and is determined to be non-human subjects research by our university’s Institutional Review Board. The model of deployment is very much akin to a “RIPE Atlas” model, a longstanding mode for deploying measurement probes in access networks.

